Paper 3.2: RAG, PRA, and Composable Native Memory

Retrieval, Materialization, Ordering, and Positional Geometry

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Abstract

Retrieval-augmented generation (RAG) reduces the amount of external information presented to a language model, while Progressive Retrieval Attention (PRA) additionally allows selected evidence to be represented, stored, and reused as model-native memory. Earlier PRA work separated retrieval from native K/V materialization, and preliminary Paper 4.5 experiments suggest a practically important regime: a strong conventional RAG retriever or reranker can choose the evidence while PRA reuses the resulting selected context as persistent native memory. When exactly the same selected block is reused, preliminary Qwen3-4B and Qwen3-8B measurements show exact generated-output preservation together with large reductions in warm time-to-first-token. However, recomposing independently cached chunks across changing selections produced a quality regression, exposing a second problem: separately stored resources have source-local positional and contextual histories that need not match the positions and causal context induced when ordinary RAG serializes them into a new prompt.

This paper studies these issues jointly across multiple RAG and long-context QA datasets. We compare full documents in context, conventional RAG with lexical, dense, hybrid, reranked, and real vector-database indices, and PRA variants that use full or partial materialization and multiple routing modes. We then freeze selected evidence and isolate realization and composition: fresh selected-text encoding, contiguous native reuse, independently cached native fragments, RoPE composition-coordinate rebinding, and limited contextual repair. Finally, we test whether arbitrary document serialization matters by comparing D_1D_2Q with D_2D_1Q and broader resource permutations under multiple positional policies, including source-local, globally packed, resource-adjacent, rank-derived, score-derived, near-band, and randomized geometries.

On five natural MultiHop-RAG cohorts (five seeds, 30 questions each), all 600 selector-frozen text/native comparisons preserved generated output and first-step logits exactly, with zero gold-answer NLL difference. Contiguous native reuse reduced warm TTFT by 92.3% and warm total latency by 81.5% relative to re-prefilling the same selected text, but an exact ordinary prefix cache remained faster. A separate 20-turn changing-selection cohort produced no reusable exact text prefix; resource-addressed PRA reused 29.0% of selected K/V entries. That saving did not imply free composition: independently encoded resources preserved only 5–10% of outputs, and even 50% contextual repair recovered 55% at best on a ten-question diagnostic. Partial materialization reduced active K/V by 25% without lowering mean token F1 on that small cohort, although output parity fell to 50%. These results establish contiguous native transport and non-prefix reuse while locating the remaining bottleneck in cross-resource contextualization, not positional transport alone.

1 Introduction

Large language models increasingly operate over collections of documents that are too large, too dynamic, or too expensive to place directly in every prompt. Conventional RAG addresses this problem by searching an external index, selecting a small evidence set, serializing that evidence as text, and re-prefilling the model. This design has two distinct consequences. First, retrieval quality becomes a major determinant of answer quality. Second, evidence that is reused across questions is nevertheless often tokenized and prefetched through the model again unless an ordinary serving cache happens to match the new prefix.

PRA introduces a different memory boundary. Retrieved information can be stored as independently addressable resources and selected native K/V can be materialized on demand. This raises a systems opportunity—persistent reuse of evidence—but also a representation question that conventional RAG hides. A prompt such as

$$D_1\, D_2\, D_3\, Q$$

imposes a total order and a precise metric distance among documents even when D_1, D_2, D_3 are logically independent sources. If these resources are encoded separately and reused later, their original positional coordinates and contextual histories are no longer automatically equivalent to the newly serialized RAG sequence.

This paper therefore asks four linked but separable questions:

1. Which external retrieval/index strategies provide the strongest candidate evidence?
2. Given the same candidates, how do conventional RAG and PRA routing compare?
3. Given the same selected evidence, what quality–cost frontier is obtained by full text, full native, and partial PRA materialization?
4. When selected native resources are independently stored, what ordering and positional composition should they use, and how much recomputation is required to recover fresh-context behavior?

The paper deliberately allows the strongest architecture to be complementary:

$$\text{mature RAG retrieval/reranking} \rightarrow \text{PRA persistent/native realization} \rightarrow \text{LLM.}$$

PRA need not beat Elasticsearch, FAISS, Qdrant, or a strong cross-encoder reranker in order to provide value.

Contributions. We make five contributions.

1. A five-seed natural test showing exact selected-text/native transport under frozen retrieval, with ordinary prefix caching as a separate control.
2. A retrieval-index comparison spanning BM25, dense FAISS, hybrid RRF, two rerankers, Elasticsearch, and Qdrant with decomposed service latency.
3. A causal materialization ladder that contrasts score-ranked, evidence-aware, and equal-budget wrong memory while reporting active native K/V explicitly.
4. A multi-resource composition study derived from the earlier Paper 1.6 plan, separating RoPE rebinding from contextual repair and document-order sensitivity.
5. A changing-selection experiment showing non-prefix native reuse and its present fidelity cost across natural overlapping query sequences.

2 Relationship to Prior PRA Work

2.1 Paper 1.5: positional transport

Paper 1.5 established source-relative positional transport, deferred/pre-RoPE machinery, and rebinding algebra. It also separated positional fragmentation from contextualization fragmentation: changing the RoPE phase of a cached key can correct effective position, but it cannot reconstruct causal context that was absent when the hidden state was originally computed.

Paper 3.2 reuses that machinery and focuses on multi-resource composition rather than re-deriving the positional transport result.

2.2 The unexecuted Paper 1.6 plan

An earlier Paper 1.6 plan proposed "multi-frame positional geometry" for independently retrieved resources. It asked whether external resources need one global positional sequence and proposed source/global, globally packed, resource-adjacent, rank-distance, score-distance, near-band, and randomized policies, together with permutation, source-distance, distractor, and multi-resource experiments. That plan was not executed as a standalone paper. Paper 3.2 incorporates its central experiments in a practical RAG setting.

2.3 Paper 3.0: causal materialization

Paper 3.0 fixes selection and varies only the native K/V detail made active. It distinguishes evidence that justifies an answer to a reader from contextual states that are computationally sufficient for a particular frozen model, and establishes that encoding, search, and materialization granularities need not coincide. Its logical interval abstraction unions overlap only within one source domain, gathers across physical storage shards, and records requested, deduplicated, evidence, non-evidence, and transferred K/V separately.

Those experiments provide two constraints for the present study. First, whole-parent disclosure is a physical control, not an oracle optimum: in the controlled study, exact cores averaged 5.25 K/V tokens and reduced K/V by 92.9% relative to whole selected parents. They raised held-out answer margin by 4.107 and reached .720 accuracy, compared with a .981 margin gain and .300 accuracy for whole parents. Equal-size wrong cores instead lowered margin by .798. The matched-content intervention shows that the effect cannot be attributed to memory size alone; broader disclosure can introduce genuine attention competition rather than harmless context. Second, a small global budget is not automatically sufficient. Under oracle selection, exact disclosure reduced active K/V by 20.9% on MuSiQue and 44.4% on 2Wiki relative to whole parents, but a common 128-token allocation did not preserve all distributed MuSiQue evidence. A later Paper 3.0 audit found that this inherited pretrained replay restarted direct-query RoPE positions at zero. The comparisons remain paired within that historical transport, but Paper 3.2 does not treat their absolute quality as corrected-position evidence. These are inherited materialization findings, not Paper 3.2 RAG results. We reuse the mechanism and causal controls, then remeasure the frontier after realistic retrieval rather than oracle selection.

Paper 3.0 also found that consumer-layer placement materially changes the effect of disclosed memory, whereas contextual representation change by itself did not explain the controlled quality gap. Paper 3.2 therefore treats retrieval identity, interval disclosure, position transport, and consumer-layer use as separate factors. A RAG chunk containing the right words is not assumed to be computationally sufficient, and a compact K/V interval is not assumed to be useful merely because its evidence density is high.

2.4 Paper 3.1: retrieval addresses

Paper 3.1 studies compact persistent retrieval indices, especially generated summaries versus lexical, extractive, and native-QK alternatives. Its conclusion is kept separate. Paper 3.2 may reuse those indices as routing arms, but the present question is end-to-end RAG composition rather than summary generation.

2.5 Paper 4.5: preliminary RAG/runtime evidence

Paper 4.5 provides the initial RAG harness, candidate and selection receipts, Selected Context and Native Memory modes, and preliminary warm-reuse measurements. The present paper treats these as motivating evidence and reruns the decisive comparisons under a frozen Paper 3.2 protocol.

3 Preliminary Evidence from Paper 4.5

The preliminary measurements already suggest that two claims should be kept separate.

Strong RAG selection plus contiguous Native Memory. A strong conventional reranker selected chunks that were then consumed either as Selected Context or as PRA Native Memory. With selection receipts frozen, the Selected Context and corresponding PRA path preserved generated outputs. Reusing the same contiguous selected native block also preserved outputs.

Representative warm measurements are summarized in Table 1. These values are inherited preliminary evidence and are not yet final Paper 3.2 results.

Table 1: Preliminary Paper 4.5 warm repeated-selection results.

Model / regime	Condition	Total latency (s)	TTFT (s)
Qwen3-4B, representative	Selected Context	3.031	2.640
	Native Memory	0.638	0.166
Qwen3-4B, 4K	Selected Context	6.121	–
	Native Memory	0.799	–
Qwen3-8B	Selected Context	5.541	4.908
	Native Memory	0.990	0.276

Persistent independent chunks. A second experiment allowed selections to change and attempted reuse of independently cached chunks. The preliminary run obtained roughly 22% mean native chunk hits and reduced 101,294 selected tokens to 80,117 uniquely materialized native tokens, with cumulative runtime decreasing from approximately 151.19 s to 131.78 s. However, the official score changed from approximately 0.440 for Selected Context to 0.380 for independently cached persistent native chunks.

This distinction motivates the central composition experiment:

$$\begin{aligned}&\text{exact same contiguous block reuse}\\ &\qquad \neq \text{arbitrary recomposition of independently contextualized chunks.}\end{aligned}$$

3.1 Inherited protocol and selection boundary

The Paper 4.5 protocol used 50 fixed MultiHop-RAG questions (seed 11) from a 609-document corpus. BM25 produced immutable candidate receipts at $N \in \{5, 10, 20, 50\}$; a 256-token chunker with 32-token overlap then supplied either global BM25 packing or parameter-free PRA rank fusion. A broad probe measured answer-string availability in selected context, not generated-answer quality. At the 2K budget and $N = 20$, PRA increased availability from .56 to .68, supporting-document coverage from .430 to .493, and span coverage from .268 to .318, while false selected-document fraction rose from .769 to .777. At $N = 50$, availability increased from .64 to .76, but support coverage changed from .515 to .505, span coverage from .338 to .300, and false-document fraction from .776 to .795. The result is a budget-sensitive selection signal, not evidence of better grounded generation.

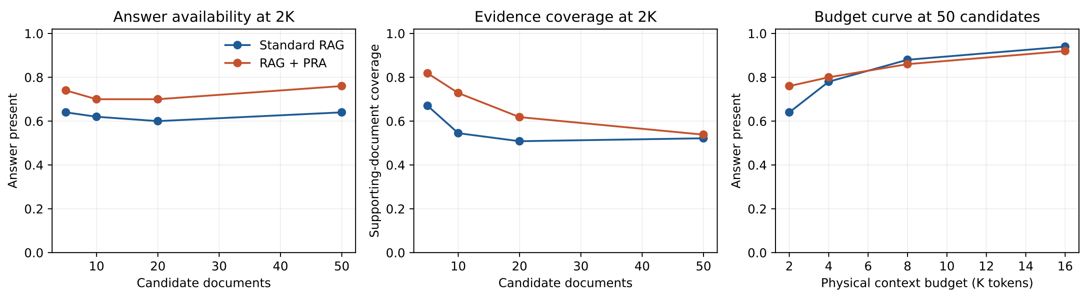

Figure 1: Inherited Paper 4.5 selection probe. The vertical measure is whether selected context contains an accepted answer string; it is not model accuracy. The source artifact remains preliminary for Paper 3.2.

3.2 Inherited powered decomposition

The powered 4B run used the same 50 questions, $N = 20$, a 2K budget, pinned Qwen3-4B 4-bit weights, and a fixed cross-encoder. Standard BM25 obtained .500 official score and .082 token F1. The stronger conventional selector obtained .440/.081, and generic PRA obtained .440/.071. The generic selector nevertheless increased supporting-document coverage from .400 to .468 and gold-chunk recall from .0448 to .0484. Thus evidence exposure and generated-answer quality moved in different directions, closing the inherited selection gate.

The matched realization result was positive and narrower. Strong conventional and strong-PRA Selected Context produced identical outputs; within generic and strong PRA, Selected Context and Native Memory also matched all generated outputs. Generic cold native execution cost 3.121 s versus 3.024 s for selected text, a 3.2% regression. Reusing the exact native selection reduced mean TTFT from 2.640 to 0.166 s and total latency from 3.031 to 0.638 s, while carrying approximately 299 MB of active all-layer K/V. The 4K boundary retained exact realization, paid a 3.3% cold regression, and reduced warm total latency from 6.121 to 0.799 s at roughly 602 MB active K/V.

The prespecified 8B replication preserved the distinction. Standard BM25, the stronger conventional selector, and generic PRA scored .520/.093, .480/.096, and .440/.081, respectively. Both native realization pairs remained output-exact. Cold native execution was 1.8% slower (5.636 versus 5.535 s), whereas retained native reuse reduced total latency from 5.541 to 0.990 s and TTFT from 4.908 to 0.276 s, using approximately 285 MB of active detail.

Finally, the independent-chunk run changed both geometry and reuse unit. It reused 22.0% of chunk lookups, reduced approximately 101,294 selected tokens to 80,117 uniquely materialized

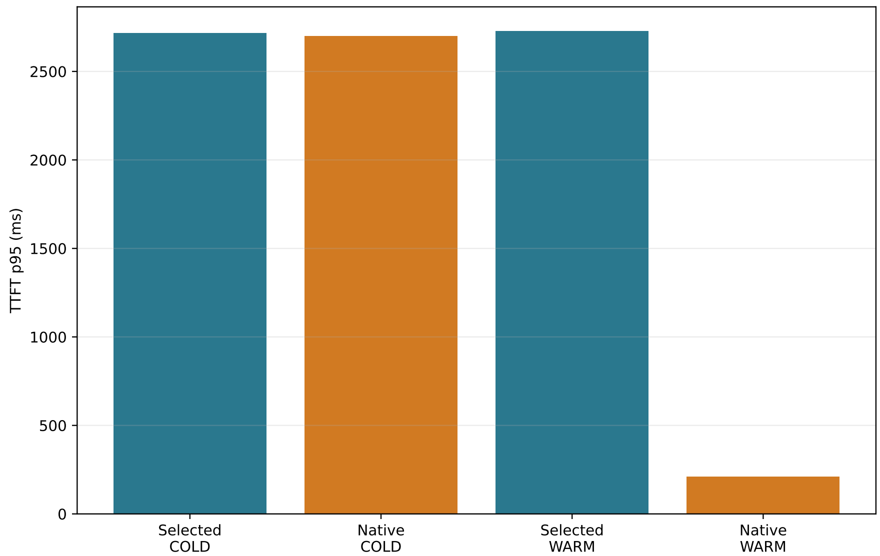

Figure 2: Inherited selector-frozen Qwen3-4B TTFT. Cold and retained-selection regimes are disjoint; the warm result is a repeated-selection benefit, not an unconditional RAG speedup or a comparison against an ordinary prefix-cache hit.

tokens, and lowered cumulative wall time from 151.19 to 131.78 s. Official score changed from .440 to .380 while token F1 changed from .071 to .086. Independently encoded chunks retained source-local post-RoPE positions instead of the fresh selected text's packed global positions. Paper 3.2 therefore treats this row as the motivating composition failure, not as transport parity or cache corruption. All numbers in this section are inherited from commit `f60b6847eb65f8a5be9c758a34a2ebdc09244e6e` and remain preliminary until reproduced here.

4 Paper 3.2 Mechanism Gate

Before loading a model, we ran the complete receipt and geometry pipeline on 15 controlled multi-document questions under five corpus seeds. Three first-stage backends produced 225 immutable candidate receipts; external-order and PRA-refined selections produced 450 selection receipts; six composed profiles and seven applicable position policies produced 4,050 composition receipts. Every receipt round-tripped with a stable digest, and all position policies preserved resource identity, source hash, token count, and internal token offsets.

At document Recall@10, BM25, exact hashed-dense retrieval, and RRF hybrid obtained .833, .900, and .913. Their external-order selected-document recalls were .587, .667, and .733. PRA second-stage ranking reached .733 for all three candidate sets. Figure 3 shows the decomposition. The synthetic fixture is deliberately lexical and small, so these numbers validate information flow and candidate ceilings; they do not rank production retrievers.

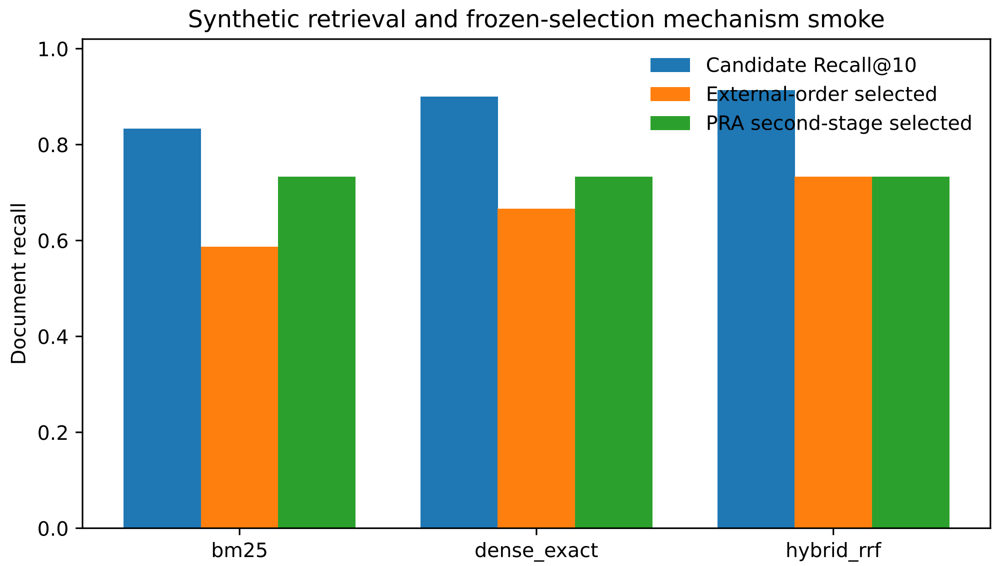

Figure 3: Five-seed synthetic retrieval and frozen-selection mechanism gate. No language model is run, and answer-quality metrics are intentionally absent.

The coordinate check also distinguishes overlap from identity collision. Globally packed, rank-distance, random-distance, and non-overlapping near-band policies assign disjoint effective positions. Source-local and resource-adjacent policies intentionally overlap approximately .807 of numeric coordinates on this fixture; score-distance overlaps .455. In every case the resource URI and chunk identity remain part of the native address, so equal position numbers from different documents are not deduplicated. This gate qualifies the experiment machinery only. Model-backed permutation, rebinding, and repair results are reported in the natural composition study below.

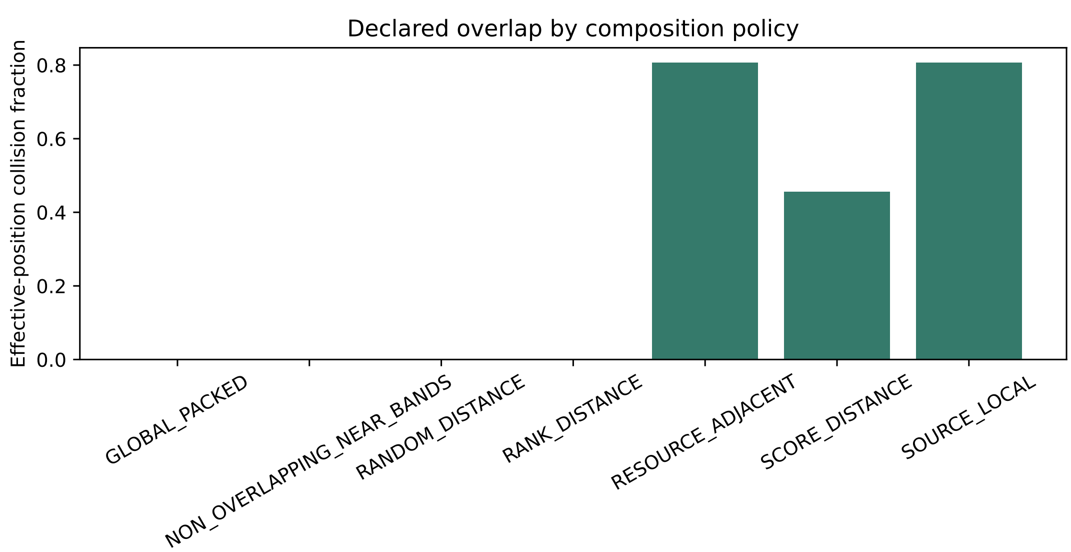

Figure 4: Effective-coordinate overlap by declared composition policy. Numeric position overlap is intentional for source-local and adjacent-resource frames; logical resource identities remain distinct.

4.1 First model-backed transport gate

We next ran the selector-frozen realization path with Qwen3-1.7B-4bit on an Apple M4 Pro. The model revision was pinned to `3b1b1768`; no PRA adaptor was available. Each of five corpus seeds used the same 15 controlled questions, while distractor construction varied by seed. We therefore use seed, rather than question execution, as the replication unit. The experiment produced 450 measured rows and 150 matched Selected Context/Native Memory pairs across cold and warm regimes.

All 150 pairs shared the same candidate receipt, selection receipt, selected intervals, and token-F1 value, and all 150 generated strings matched exactly. Table 2 reports means across the five seed summaries. Native realization replaced approximately 122 visible evidence tokens with the same number of selected native K/V states; the direct visible prompt fell from 153.6 to 32.0 tokens. Cold native execution remained 3.3% slower in TTFT and 4.1% slower end to end. Once the exact selection was retained, native ingestion fell from 60.5 to 0.04 ms, TTFT from 102.2 to 37.0 ms, and total latency from 177.8 to 116.5 ms.

Table 2: Five-seed controlled Qwen3-1.7B transport gate. Values are means of 15-question seed summaries; parentheses give between-seed standard deviation.

Regime	Realization	F1	visible	TTFT (ms)	total (ms)
Cold	Selected text	.094 (.010)	153.6	100.6 (1.7)	176.2 (2.7)
Cold	Native K/V	.094 (.010)	32.0	103.9 (2.5)	183.5 (2.7)
Warm	Selected text	.094 (.010)	153.6	102.2 (1.8)	177.8 (2.5)
Warm	Native K/V	.094 (.010)	32.0	37.0 (0.4)	116.5 (0.5)

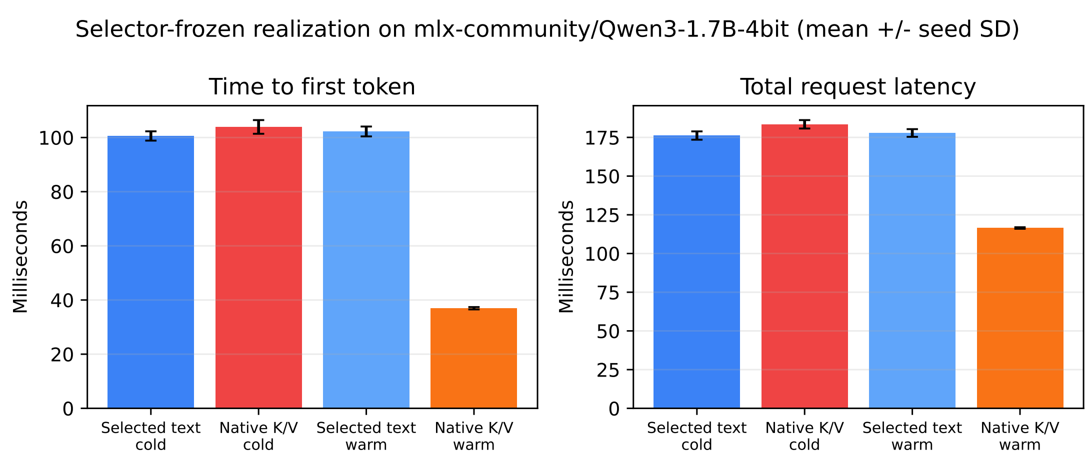

Figure 5: Matched realization cost on the controlled fixture. Error bars are between-seed standard deviations. Native reuse is beneficial only in the warm regime here; the cold path pays native encoding and is slightly slower.

This result closes only the narrow transport gate. Standard BM25 achieved mean token F1 .109 and supporting-document coverage .933, while the generic PRA selector obtained .094 and .873. The fixture is lexical, small, and repeated across seeds; exact match was zero for every arm because the model returned sentences rather than the requested short value. Accordingly, the selection gate remains closed on this fixture.

4.1.1 Natural MultiHop-RAG replication

We then ran the same pinned model on 50 sampled MultiHop-RAG questions at the inherited $N = 20$, 2K-token operating point, with a cross-encoder pinned to revision `233902d2`. This is one fixed cohort (seed 11), not five independent natural cohorts. All 200 cold/warm Selected Context/Native Memory pairs—100 generic and 100 cross-encoder-selected—again shared candidate and selection receipts, intervals, scores, and exact generated output. Table 3 separates selection from realization.

Table 3: Qwen3-1.7B on 50 MultiHop-RAG questions. "Vis./native" reports direct visible prompt tokens and selected native K/V tokens.

Condition	Regime	official	F1	support	vis./native	TTFT	total
BM25 text	Cold	.480	.119	.400	2108/0	1013	1105
Cross-encoder text	Cold	.500	.137	.455	2110/0	1020	1111
PRA generic selected	Cold	.440	.122	.468	2105/0	1000	1092
PRA strong selected	Cold	.500	.137	.455	2110/0	1003	1095
PRA strong native	Cold	.500	.137	.455	79/2031	1005	1129
PRA strong native	Warm	.500	.137	.455	79/2031	73	196

Generic PRA raised answer-string availability from .56 to .64 and supporting-document coverage from .400 to .468; gold-chunk recall moved from .0448 to .0484. The official score nevertheless fell from .48 to .44, while token F1 changed from .119 to .122. The cross-encoder was stronger: answer availability .70, official score .50, and token F1 .137. Its ordinary RAG, PRA Selected Context, and PRA Native Memory outputs agree exactly. The result therefore supports conventional retrieval followed by PRA realization, but does not establish PRA as a better first- or second-stage selector.

Cold strong-selector native execution was 3.1% slower end to end than its matched Selected Context arm, while TTFT changed by only 0.2%. Retaining the exact selection reduced TTFT from 999 to 73 ms (92.7%) and total latency from 1091 to 196 ms (82.0%), at 222.1 MiB of active all-layer K/V.

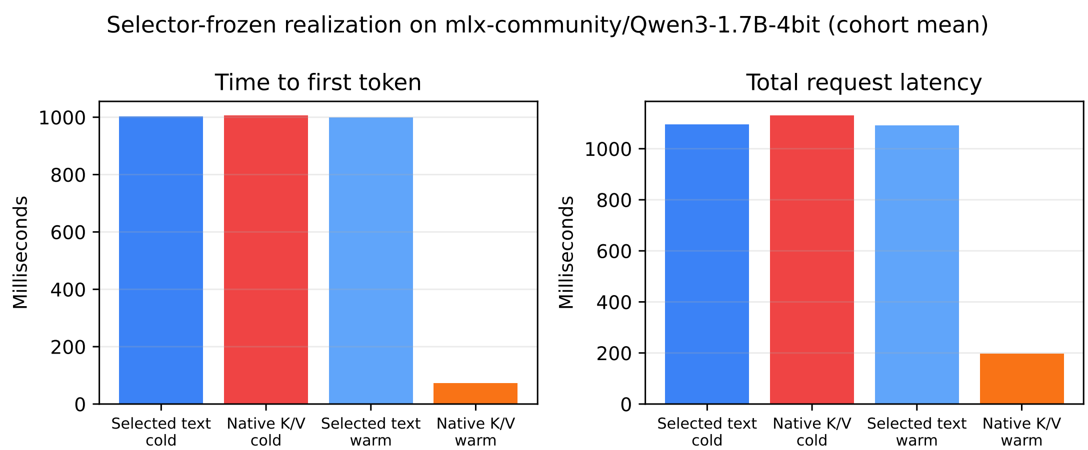

Figure 6: Single-cohort MultiHop-RAG realization cost with selector output frozen. No between-seed error bars are available for this first natural run.

The native and repeated-selection economic gates pass at the prespecified 50-example threshold. The selection gate fails because generic PRA does not match the strongest conventional selector.

The public-card gate also remains closed because no exact model/precision adaptor is qualified. The five-seed answer-log-probability and independently composed-resource experiments below extend this initial cohort; cross-family replication remains required.

5 Experimental Questions

5.1 Full context, RAG, or PRA?

When all candidate documents fit the native context window, we compare:

1. **Full-Doc-In-Context**: all candidate documents are freshly encoded;
2. **conventional RAG**: retrieved chunks are freshly packed as text;
3. **PRA full materialization**: PRA routes but exposes all tokens of selected resources;
4. **PRA partial materialization**: PRA exposes a bounded selected subset;
5. **PRA Native Memory**: selected evidence is represented by reusable native K/V.

Full context is a reference rather than a guaranteed upper bound: retrieval may improve quality by removing distractors.

5.2 Does PRA need to replace the RAG retriever?

No such assumption is made. We test:

$$\begin{aligned}
&\text{external retrieval} \rightarrow \text{ordinary RAG},\\
&\text{external retrieval} \rightarrow \text{PRA second-stage selection},\\
&\text{strong reranker} \rightarrow \text{PRA Selected Context},\\
&\text{strong reranker} \rightarrow \text{PRA Native Memory}.
\end{aligned}$$

This explicitly separates selection quality from realization and reuse.

5.3 RAG+PRA as a composed system

We represent RAG+PRA as a sequence of independently auditable stages:

$$\begin{aligned}
&\text{query and corpus revision} \rightarrow \text{external retrieval/index} \rightarrow \text{candidate receipt}\\
&\rightarrow \text{optional PRA second-stage routing} \rightarrow \text{selection receipt} \rightarrow \text{materialization and composition}\\
&\rightarrow \text{Selected Context or Native Memory} \rightarrow \text{persistent reuse} \rightarrow \text{generation.}
\end{aligned}$$

External retrieval may use BM25, dense/FAISS search, hybrid fusion, a strong reranker, or a retrieval service. Its immutable candidate receipt records source identities, spans, scores, ranks, and index revision. The subsequent selection receipt fixes the exact evidence consumed by matched realization conditions. In particular, a native-memory condition cannot rerun retrieval and still be described as a transport comparison.

Table 4 defines the composed profiles used throughout the study. These profiles let us test PRA in three distinct roles: replacement selector, second-stage selector, and realization/reuse layer after a conventional selector.

The first comparison is always `RAG_ONLY_TEXT` against the realization-only native profiles with identical evidence. We then evaluate external retrieval followed by PRA refinement, and only afterward PRA as a replacement selector. Thus a retrieval improvement, a materialization saving, and a positional-recomposition result remain separate empirical claims.

Table 4: Canonical RAG+PRA profiles. Conditions sharing a selection receipt hold evidence fixed and change only its realization or composition.

Profile	Selection	Realization and purpose
`RAG_ONLY_TEXT`	External RAG	Freshly packed visible text; strong conventional baseline.
`RAG_PLUS_PRA_SELECTED`	External RAG, optionally PRA-refined	Fresh Selected Context; isolates second-stage routing without native transport.
`RAG_PLUS_PRA NATIVE_CONTIGUOUS`	Frozen external selection	One reusable packed native block; isolates reuse under ordinary packed geometry.
`RAG_PLUS_PRA NATIVE_INDEPENDENT`	Frozen external selection	Independently cached source-position K/V; exposes composition error.
`RAG_PLUS_PRA_NATIVE_REBOUND`	Frozen external selection	Independent K/V rebound to request-specific composition positions; isolates positional repair.
`RAG_PLUS_PRA_REPAIR`	Frozen external selection	Rebound K/V plus bounded contextual repair; estimates the recomputation needed for fresh-context parity.

6 Datasets

The study uses a staged dataset ladder.

Mechanism and inherited QA. Controlled synthetic multi-document facts provide exact causal checks. Natural PRA cohorts include HotpotQA, QASPER, 2WikiMultiHopQA, and MuSiQue.

Flagship RAG. MultiHop-RAG is the mandatory first flagship dataset because preliminary Paper 4.5 infrastructure already exists. Once the protocol is stable we add KILT Natural Questions and KILT HotpotQA.

For each dataset we freeze corpus and query revisions, official metrics, chunking, candidate document construction, evidence labels where available, and the subset for which Full-Doc-In-Context fits the model's native window.

7 Model Scaling

The full experimental grid is intentionally run only on small models.

Stage M0. Qwen3-0.6B is the primary mechanism model. Llama 3.2-1B provides an early cross-family check where practical.

Stage M1. After correctness gates, a 1–2B class model is used for reduced replication.

Stage M2. Qwen3-4B is mandatory because it connects directly to Paper 4.5 preliminary RAG evidence. Qwen3-8B is the first scaling replication.

Stage M3. Only the strongest conditions are repeated on Qwen3-14B and, if compute permits, a larger Qwen and at least one non-Qwen family.

This staged design tests whether an apparent mechanism survives scale without multiplying every retrieval, position, and materialization factor at every model size.

8 Retrieval and Index Strategies

8.1 Local/custom indices

We begin with transparent in-process baselines.

BM25. A pinned BM25 implementation provides the lexical reference.

Dense semantic retrieval. The requested "FAS" path is interpreted as FAISS. We use one frozen embedding model and compare exact dense search on small corpora with a pinned FAISS ANN configuration on larger corpora.

Hybrid. Lexical and dense candidates are fused first using reciprocal-rank fusion (RRF). A predeclared weighted fusion may be added as a secondary arm.

Strong conventional reranker. A frozen cross-encoder or equivalently strong conventional reranker provides both a competitive RAG baseline and a selector-frozen input to PRA realization.

8.2 Real retrieval services

The remote machine will host containerized services. The initial target set is:

- Elasticsearch (or a precisely pinned OpenSearch-compatible path where explicitly intended);
- Qdrant;
- one of Weaviate or Milvus;
- pgvector where operationally inexpensive.

The user-specified "Derby" backend is retained as a requested integration, but its exact product and API must be identified before implementation; the experiment must not silently assume that Apache Derby supplies vector ANN semantics.

The purpose is not to publish a standalone vector-database benchmark. Instead we ask whether RAG/PRA conclusions survive realistic retrieval infrastructure. Service latency is decomposed into search, network, reranking, and LLM costs.

9 PRA Routing and Materialization

9.1 Routing families

Within an identical candidate corpus, candidate PRA routing modes include:

- lexical routing;

- native semantic/gist routing;
- native-QK or the strongest inherited low-rank QK selector where available;
- lexical–semantic hybrid routing;
- the strongest inherited generic PRA profile;
- optional Paper 3.1 summary indexing as a secondary arm.

9.2 Materialization families

We distinguish:

Full. Materialize all tokens from each PRA-selected object.

Fixed partial. Materialize a fixed chunk/token budget.

Score partial. Materialize under a frozen threshold/budget policy.

Hierarchical. Use inherited hierarchical routing/materialization where supported.

Selected Context. Render exactly the selected evidence as freshly encoded ordinary text.

Native contiguous. Encode the selected evidence once as one contiguous native block and reuse it.

Native independent. Reuse independently cached chunks/resources and compose them at request time.

Every condition reports candidate, selected, visible, native, newly materialized, and reused token counts. Following Paper 3.0, partial conditions operate on source-relative logical intervals. Overlap is deduplicated only within one resource domain, so equal numeric offsets in two documents remain distinct native states. In addition to the counts above, each run reports requested tokens before union, unique native tokens, evidence and non-evidence tokens, evidence density, storage shards touched, and cross-shard intervals. Whole-parent and equal-budget wrong-memory conditions act as causal controls where evidence labels permit them. Any fixed budget that omits a required evidence region is reported as a coverage failure, not as evidence that compact materialization is sufficient.

10 Matched Selection and Realization

The most important causal control freezes the selected chunks. For exactly the same selection receipt we compare:

1. fresh selected text;
2. contiguous native reuse;
3. independently cached native fragments at source-local positions;
4. the same independent fragments rebound to packed composition positions;
5. rebound fragments with optional limited contextual repair.

This design ensures that a quality difference after selection cannot be mislabeled as a retrieval failure.

11 Multi-Resource Positional Geometry

11.1 Source and composition coordinates

For resource r and token i, distinguish

$$p_{\text{src}}(r,i) \qquad \text{from} \qquad p_{\text{cmp}}(r,i).$$

The first identifies position in the source's own positional frame. The second is the effective position chosen for the current materialized composition.

A resource-preserving offset can be written

$$p'_{r,i} = b_r + p_{r,i}.$$

The value of b_r is a policy choice, not a consequence of physical K/V residency.

11.2 Required policies

SOURCE_LOCAL. Preserve each resource's source-local coordinates.

GLOBAL_PACKED. Assign selected resources contiguous positions exactly as in freshly packed RAG. This is the primary parity policy.

RESOURCE_ADJACENT. Preserve each resource's internal geometry while independently placing its end near the query. Different resources may overlap in effective RoPE coordinate ranges while remaining separate K/V objects.

RANK_DISTANCE. Map retrieval rank to query distance.

SCORE_DISTANCE. Map a normalized retrieval score to query distance under a frozen mapping.

NON_OVERLAPPING_NEAR_BANDS. Use bounded near-query bands while preserving a monotonic resource ordering.

RANDOM_DISTANCE. Randomize resource offsets under a matched envelope as a negative control.

All policies use one ordinary attention normalization over the concatenated selected K/V. The experiment changes positional transforms, not the definition of attention.

12 Document-Order and Permutation Sensitivity

The minimal experiment compares

$$D_1\, D_2\, Q \qquad \text{with} \qquad D_2\, D_1\, Q.$$

For three or more resources we additionally compare canonical retrieval order, reverse order, and random permutations.

We distinguish two experiments.

Composition sensitivity. Resource identities, selected content, selected-token budget, and model are frozen. Only serialization order and/or positional policy changes. Retrieval is not rerun.

Pipeline sensitivity. Each retrieval/reranking method is allowed to return and pack its natural order. This measures complete RAG behavior.

For permutation set $\mathcal{P}$ we report answer flips, target probability variance, and mean pairwise Jensen–Shannon divergence:

$$S_{\text{order}} = \frac{2}{|\mathcal{P}|(|\mathcal{P}|-1)} \sum_{a<b} JS(P_a \| P_b).$$

A compelling multi-frame result requires maintained or improved task quality in addition to reduced arbitrary permutation variance.

13 Source Distance, Distractors, and Number of Resources

Following the earlier Paper 1.6 plan, source-distance experiments hold evidence fixed while varying its source/global distance from the query. Distractor experiments sweep $0, 2, 4, 8, 16, 32$ irrelevant resources where context permits. Relevant-resource experiments sweep $1, 2, 4, 8$ sources and include redundant, complementary multi-hop, and conflicting evidence.

These controls are especially important for RESOURCE_ADJACENT: placing every resource near the query may reduce lost-in-the-middle effects but could also make irrelevant evidence excessively salient.

14 Positional vs Contextualization Fragmentation

Consider a selected composition A, B, C, Q.

C0: Fresh packed. Freshly encode $[A\,B\,C\,Q]$. This establishes the matched-content reference.

C1: Native source. Reuse independently cached native A, B, C at their source-local effective positions.

C2: Native rebound packed. Keep the independently computed hidden/K/V states but rebind RoPE keys to the positions that A, B, C would occupy in the packed composition.

For pre-RoPE keys,

$$K_{\text{cmp}} = R_{p_{\text{cmp}}} K_{\text{raw}}.$$

For validated post-RoPE storage,

$$K_{\text{cmp}} = R_{p_{\text{cmp}} - p_{\text{src}}} K_{\text{stored}},$$

subject to the exact model's RoPE/scaling implementation.

C3: Rebound plus repair. If positional rebinding does not restore sufficient quality, selectively recompute a fraction of the selected context.

C4: Full recompute. Fresh reprefill provides the full contextualization reference.

A divergence D can summarize the fraction of source-position error recovered by rebinding:

$$R_{\text{pos}} = 1 - \frac{D(P_{\text{fresh}}, P_{\text{rebound}})}{D(P_{\text{fresh}}, P_{\text{source}})}.$$

The residual gap is not automatically purely contextual, but it directly bounds what positional phase correction alone can achieve.

15 Selective Contextual Repair

If C2 remains measurably different from fresh encoding, we test a deliberately simple repair ladder:

$$0\%,\ 5\%,\ 10\%,\ 25\%,\ 50\%,\ 100\%$$

of selected tokens or equivalent compute, together with boundary-window and first-N-tokens-per-resource controls.

A useful result is a Pareto curve:

$$\text{quality/parity recovered} \quad \text{vs} \quad \text{fraction recomputed} \quad \text{vs} \quad \text{TTFT/prefill}.$$

Adaptive repair is postponed until fixed policies establish headroom.

16 Repeated-Query RAG and Persistent Reuse

Real sessions rarely repeat exactly one selected prompt. We therefore construct query sequences such as

$$Q_1 \to \{A, B, C\}, \qquad Q_2 \to \{B, C, D\}, \qquad Q_3 \to \{A, D, E\}.$$

We compare:

1. conventional RAG with full reprefill;
2. conventional engine prefix/KV caching;
3. PRA exact contiguous-block reuse;
4. PRA independent native reuse at source positions;
5. PRA independent native reuse with packed rebinding;
6. packed rebinding plus repair.

We report selection overlap, native chunk-hit fraction, newly encoded and reused tokens, cumulative TTFT, cumulative prefill, wall time, memory residency, official quality, and output flips.

This is the main product-facing experiment: can PRA reuse semantically recurring, non-prefix evidence while an ordinary RAG retriever continues to determine what is relevant?

17 Ordinary Cache Controls

The preliminary warm Native Memory result is not sufficient by itself. On at least one production-style engine we compare:

RAG without cache,
RAG + ordinary prefix/APC cache,
RAG + PRA Native Memory,
RAG + prefix cache + PRA.

Exact-prefix cache hits and query-addressed PRA resource reuse are reported independently.

18 Metrics and Statistics

18.1 Quality

We use official EM/F1/accuracy where defined, gold NLL/log-probability, answer margin, generated sequence agreement, and a frozen semantic judge only where official metrics are insufficient.

18.2 Retrieval

Document Recall@k, supporting-document/span coverage, MRR, nDCG where meaningful, candidate counts, and reranker displacement are reported.

18.3 Context and memory

Metrics include logical candidate tokens, selected/visible/native/new/reused tokens, materialization avoidance, active K/V bytes, and selected/full ratio.

18.4 Composition

We report answer flips, pairwise JS/KL, target-probability variance, source-distance slopes, distractor sensitivity, and fresh/source/rebound divergences.

18.5 Systems

We report index build time, retrieval p50/p95/p99, rerank latency, prefill, TTFT p50/p95/p99, ITL, completion latency, tokens/s, requests/s, memory, K/V bytes, transfer traffic, and cold/warm/hot state.

All matched conditions use paired statistics and bootstrap 95% confidence intervals. Model- and workload-specific sign reversals remain visible rather than being averaged away.

19 Experimental Schedule

19.1 Phase 0: audit and inheritance

Paper 4.5 artifacts are copied with provenance and labeled preliminary. Dataset, model, retriever, chunker, and generation revisions are verified.

19.2 Phase 1: small-model baseline

Qwen3-0.6B runs Full-Doc-In-Context, BM25, FAISS dense, hybrid, strong-reranker RAG, and Full Selected Context. Candidate and selection receipts are frozen in this phase. The first composed comparison runs `RAG_ONLY_TEXT` against `RAG_PLUS_PRA_NATIVE_CONTIGUOUS`, isolating representation and reuse before changing the selector.

19.3 Phase 2: PRA routing/materialization

On identical candidate receipts we run the routing families and full/partial materialization ladder. Clearly dominated arms are pruned. The complete RAG+PRA profile matrix is then crossed with conventional selection, PRA second-stage selection, and PRA replacement selection. Realization arms within each selector condition share one immutable selection receipt.

19.4 Phase 3: ordering and position

The seven positional policies, D_1D_2/D_2D_1, broader permutations, source distance, and distractors are run on the small model.

19.5 Phase 4: composition fidelity

Fresh packed, source-native, packed-rebound, and repair conditions isolate positional and contextualization effects.

19.6 Phase 5: real retrieval services

Elasticsearch and at least one dedicated vector database are deployed remotely. Qdrant, Weaviate, Milvus, and pgvector integrations are added as budget permits. The requested Derby integration is added after exact backend identity is pinned.

19.7 Phase 6: powered 4B study

Qwen3-4B repeats only frozen best conditions on MultiHop-RAG and key inherited QA cohorts. The preliminary Paper 4.5 rows are reproduced or explicitly marked non-reproduced.

19.8 Phase 7: 8B replication

Qwen3-8B runs the decisive reduced matrix.

19.9 Phase 8: larger/cross-family

Qwen3-14B and one non-Qwen model provide minimal scaling/family confirmation when compute permits.

19.10 Phase 9: KILT validation

KILT Natural Questions and KILT HotpotQA are added only after the protocol is stable.

20 Measured Retrieval and Native-Memory Results

20.1 Retrieval remains an independent component

Table 5 reports a frozen 50-question MultiHop-RAG retrieval cohort. The hybrid and stronger reranker results make clear why the native-memory experiment should not silently substitute a weaker selector. The mean reranker timings include first-use costs and are not steady-state serving estimates.

Table 5: Local CUDA retrieval at $K = 20$. "All" requires every supporting document; "mean" averages per-question supporting-document recall.

Method	All support	Mean support	Mean latency (ms)
BM25	.54	.735	7.1
FAISS + BGE dense	.58	.803	12.3
Historical MiniLM reranker	.58	.815	371.9
Hybrid RRF	.70	.870	20.3
BGE-v2-M3 reranker	.78	.897	3868.6

Elasticsearch, Qdrant, and their service-level RRF hybrid reached all-support recall@20 of .56, .52, and .76, respectively. On the RTX 5060 run, mean Elasticsearch service search was 11.46 ms. Qdrant search itself was 2.60 ms; its 23.41 ms end-to-end value included 20.76 ms of BGE query embedding. The hybrid path measured 53.12 ms end to end, including 11.32 ms embedding and 40.75 ms service work. Figure 7 presents the matched local comparison. These results evaluate retrieval, not PRA transport.

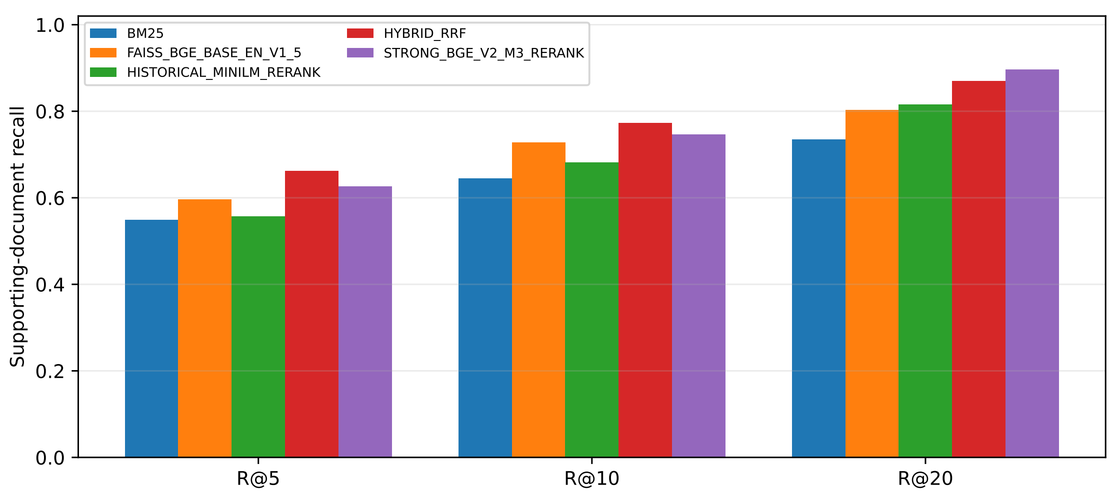

Figure 7: Supporting-document recall for the local retrieval ladder. All methods see the same 50 questions and corpus revision.

20.2 Natural composition and contextual repair

We next freeze the historical cross-encoder's selection for ten natural questions and four resource orders per question. Table 6 compares each native realization with fresh packed encoding. Contiguous reuse is numerically exact in all 40 comparisons. Independent source-local encoding is not: changing only the RoPE phase to the fresh packed coordinates improves mean JS and NLL error but does not restore the missing cross-resource causal context. At 50% recomputation, repairing a contiguous prefix of later resources is more faithful than repairing boundaries, but both recover only 22 of 40 generated outputs. Full recomputation is required for exact parity in this cohort.

Table 6: Natural independent-composition diagnostic (40 order-specific comparisons). NLL is mean absolute gold-answer NLL error versus fresh packed.

Realization	Output parity	JS	$\lvert\Delta\text{NLL}\rvert$
Contiguous native block	40/40	.000	.000
Independent source-local	2/40	.179	2.143
Global-position rebound	3/40	.149	1.946
Boundary repair, 25%	14/40	.077	1.166
Boundary repair, 50%	22/40	.052	.781
Later-resource prefix repair, 50%	22/40	.024	.468
Full repair	40/40	.000	.000

Figure 8: Position rebinding and deterministic contextual repair reduce the composition error, but this cohort reaches exact parity only at full repair. Repair timings are diagnostic because fresh packed states were computed first.

Serialization itself is consequential: nine of ten questions generated two or three distinct outputs across canonical, reverse, and two random orders. Gold NLL variance ranged from zero to 2.90. This establishes order sensitivity but does not yet identify an alternative position policy that preserves quality.

20.3 Alternative positional geometries

A reduced two-order sweep evaluates the Paper 1.6 position policies on the same ten questions. None dominates packed-position rebinding. GLOBAL_PACKED gives the lowest mean JS divergence (.152) and absolute NLL error (1.947) among the reusable geometries, but preserves only one of 20 outputs. Near bands, rank distance, and random distance each preserve three outputs while increasing JS to .179–.188. RESOURCE_ADJACENT is weakest, with zero output matches, .216

JS, and 2.283 NLL error. Position geometry changes the failure, but does not replace missing cross-resource contextualization.

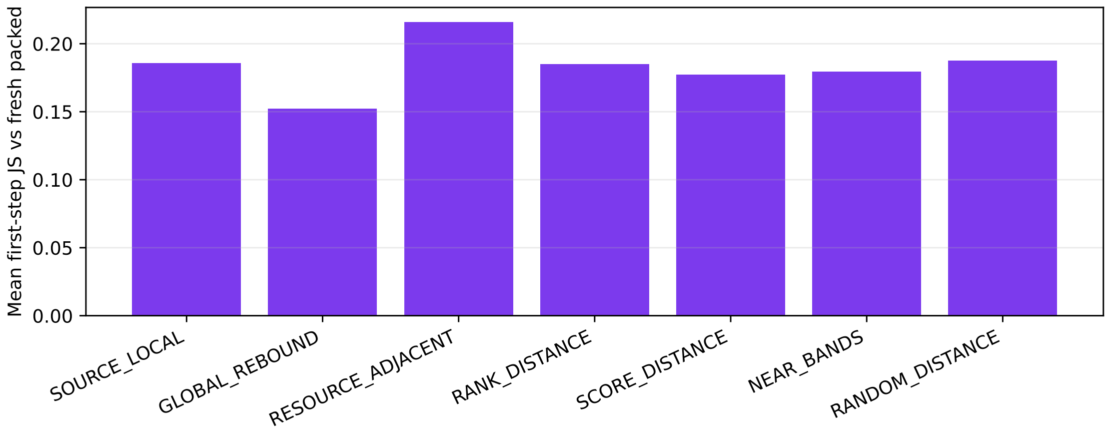

Figure 9: First-step divergence from fresh packed encoding under alternative position policies (20 order-specific comparisons per policy).

20.4 Partial materialization

Table 7 holds selected resources and order fixed, then reduces active native K/V. Score-ranked 75% materialization kept the cohort's mean token F1 at .213 while lowering active K/V from 1325.5 to 994.1 and TTFT from 91.9 to 69.0 ms. It preserved only five of ten generated outputs, however, and smaller fractions were not monotonic. The equal-budget wrong-memory control reduced evidence coverage to .05 and token F1 to .08. This ten-question result is a mechanism signal, not a calibrated deployment policy.

Table 7: Canonical-order partial-materialization diagnostic.

Policy	Active K/V	Support	F1	NLL	TTFT (ms)
Full	1325.5	.467	.213	4.376	91.9
Score, 75%	994.1	.442	.213	3.656	69.0
Score, 50%	662.8	.333	.073	3.911	66.6
Score, 25%	331.4	.283	.120	3.806	62.1
Score, 12.5%	165.7	.283	.080	4.937	62.8
Evidence oracle, 25%	331.4	.358	.145	3.753	63.4
Wrong memory, 25%	331.4	.050	.080	5.190	62.9

20.5 Changing-selection reuse

The product-relevant test groups 20 natural questions into five four-turn sequences chosen to overlap selected resources. Ordinary exact-prefix caching reused zero evidence tokens because each question changed the serialization. Resource-addressed PRA reused 384.2 of 1326.2 selected K/V entries per turn on average (29.0%), lowering measured total time from 720.8 to 570.1 ms. Savings varied substantially by sequence, from 11.9% to 57.0% fewer newly encoded entries.

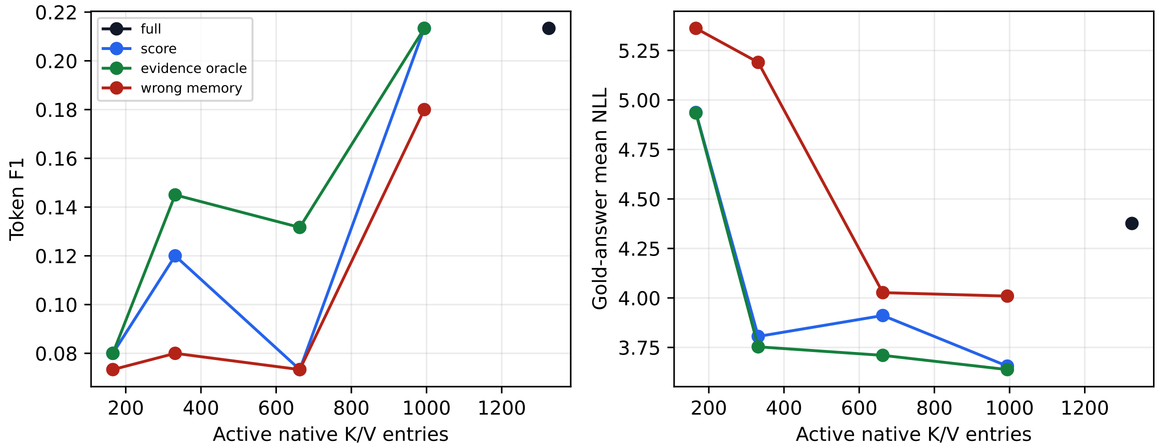

Figure 10: Quality and gold-answer NLL against active native K/V. The different metrics need not rank small-sample generations identically.

This is not an end-to-end quality win yet. Global rebound preserved two of 20 fresh outputs and 25% diagnostic repair preserved nine. Their mean token F1 was .130 and .115, respectively, versus .174 for fresh packed encoding. Partial 50% materialization lowered newly encoded K/V by 58.9% and total time by 47.6%, but preserved one of 20 outputs and reached .102 F1. Thus PRA finds reuse that an exact prefix cache cannot express, while the present independently encoded representation still needs a better contextualization mechanism.

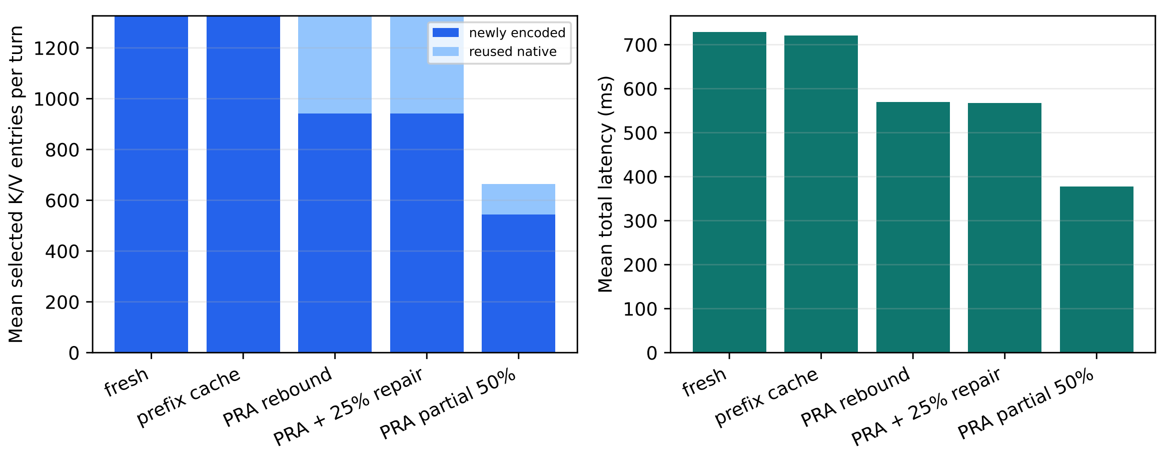

Figure 11: Mean per-turn K/V accounting and latency under changing selections. The light segment is resource-addressed native reuse; fidelity is reported in the text and must not be inferred from cost alone.

21 Interpretation and Falsification

Three claims now have different evidential status. First, contiguous native realization is established for the measured model and cohorts: selection, output, logits, and answer NLL agree. Second, non-prefix resource reuse is measurable where exact-prefix reuse is zero. Third, useful composition is not yet established: RoPE rebinding removes only part of the error, and the tested bounded repair policies do not recover fresh behavior reliably.

The results therefore falsify the strongest form of the composition hypothesis. Post-RoPE K/V is not a freely movable document representation. Later-layer states carry causal context from the serialization in which they were formed. This does not invalidate persistent native memory, but it narrows its present safe use to contiguous blocks, unchanged selections, or explicitly repaired fragments.

Partial materialization remains promising but provisional. Its 75% point matches mean F1 on ten examples, while non-monotonic smaller budgets and output flips show that evidence coverage and attention competition cannot be reduced to token count alone. A larger held-out replication must precede a product profile. Likewise, the measured order sensitivity motivates multi-frame geometry but does not count as evidence that any alternative geometry is better.

The strongest baseline boundary is also retained: for an identical repeated serialized prefix, ordinary prefix caching is faster than native PRA. PRA's distinct opportunity begins when resources recur across changed questions, orders, or evidence combinations. Even there, the current cost saving becomes useful only if representation fidelity is repaired.

22 Systems and Reproducibility

Every run persists dataset/corpus revision, query identity, candidate receipt, candidate scores, reranker scores, selected resource/chunk IDs, selected order, spans, materialization receipt, source and composition positions, position policy, model/tokenizer revision, generation settings, backend/index version, hardware, and run/commit IDs.

Real retrieval services run in pinned Docker/Compose environments. Network/service latency is separated from model inference. FAISS in-process latency is not directly compared with remote-service end-to-end latency without decomposition.

23 Discussion

The broader representational question is whether a transformer must interpret a retrieved evidence set through a fabricated one-dimensional global sequence. Local order inside a document is usually meaningful; structural order among sections can also be meaningful; but serialization order among independently retrieved resources may be arbitrary. PRA makes this distinction explicit because logical identity, physical K/V residency, source coordinates, and request-specific composition coordinates can be represented separately.

The immediate result is conservative. Rebinding independently encoded keys to ordinary packed-RAG positions improves numerical fidelity, but does not recover the contextual history of packed encoding. Resource-relative geometries should therefore be evaluated as explicit alternatives, not described as equivalent coordinate changes.

24 Conclusion

Paper 3.2 evaluates PRA beneath a conventional RAG stack rather than requiring it to replace mature retrieval. Across five natural cohorts, contiguous native K/V reproduced frozen selected-text behavior exactly and made repeated reuse substantially cheaper than text re-prefill. An exact prefix cache remained the better baseline when the serialized prefix itself repeated.

Changing selections reveal both the opportunity and the hard boundary. PRA reused 29.0% of selected K/V where ordinary prefix caching reused none, but independently encoded resources were not compositionally equivalent to a fresh packed prompt. Position rebinding helped; 50% contextual repair helped more; neither restored exact behavior reliably. Partial materialization found one promising 75% operating point on ten questions, but the smaller budgets were non-monotonic and require replication. The practical architecture is therefore clearer than the final solution: external search can freeze strong evidence, PRA can store and reuse exact contiguous native blocks, and future work must make changed compositions contextually faithful without paying full reprefill.

A Artifact Schema

Committed artifact groups include local and service-backed retrieval, five-seed matched realization, partial materialization, independent composition, repair curves, permutation receipts, and changing-selection reuse. Each group retains raw rows or compressed raw rows plus a manifest. Cross-model scale and full positional-policy artifacts use the same schema.

B Initial Service Matrix

Backend	Role	Deployment	Status
BM25 local	lexical reference	in process	measured, 50 questions
FAISS + BGE	dense reference	in process	measured, 50 questions
Elasticsearch	lexical reference	remote Docker	measured, 50 questions
Qdrant + BGE	vector reference	remote Docker	measured, 50 questions
RRF service hybrid	lexical + vector	remote Docker	measured, 50 questions
Weaviate or Milvus	vector/hybrid	remote Docker	deferred
pgvector	DB-integrated vector	remote Docker	optional/planned
Derby	user-requested backend	TBD	identity to resolve